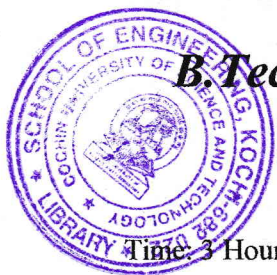


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B.Tech. Degree IV Semester Examination April 2019

SE 15-1402 HEAT AND MASS TRANSFER OPERATIONS (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Derive an expression for steady state heat conduction through a hollow cylinder.
- (b) What are the different dimensionless numbers used in case of forced convection?
- (c) Differentiate between film type and drop wise condensation.
- (d) Explain LMTD with reference to parallel and counter-current flow.
- (e) State and explain the different laws of radiation.
- (f) What is Fick's law of diffusion?
- (g) For a solution, prove that $J_A = -J_B$.
- (h) What are the factors to be considered while selecting an absorbent?
- (i) What is meant by relative volatility?
- (j) Differentiate between evaporation and drying.

PART B

(4 × 10 = 40)

- II. (a) The walls of a house in a cold region consist of three layers-outer brickwork of 15 cm thickness and an inner wooden panel of 1.2 cm thickness. The intermediate layer is made of an insulating material 7 cm thick. The thermal conductivities of the brick and the wood used are 0.70 W/m °C and 0.18 W/m °C, respectively. The inside and outside temperatures of the composite wall are 21 °C and -15 °C, respectively. If the layer of insulation offers twice the thermal resistance of the brick wall, calculate: (i) the rate of heat loss per unit area of the wall and (ii) the thermal conductivity of the insulating material. (6)
- (b) Derive an expression for the heat conduction through a series of resistances in a flat wall. (4)

OR

- III. Describe pool boiling heat transfer with a neat sketch of the hysteresis curve. (10)
- IV. (a) In a counter flow double pipe heat exchanger, oil is cooled from 85°C to 55°C by water entering at 25°C. The mass flow rate of oil is 9,800 kg/h and specific heat of oil is 2000 J/kg K. The mass flow rate of water is 8,000 kg/h and specific heat of water is 4180 J/kg K. Determine the heat exchanger area and heat transfer rate for an overall heat transfer coefficient of 280 W/m² K. (6)
- (b) What are the methods of feeding in multiple effect evaporators? (4)

OR

(P.T.O.)

- V. (a) Write a note on blackbody radiation. What is meant by shape factor? (5)
 (b) A black body at 3000 K emits radiation. Calculate the (i) wave length for which emission is maximum ($\lambda_{\max}$) and (ii) the total emissive power E_b . Stefan Boltzmann constant = $5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$. (5)

- VI. Explain the design concepts of a packed tower for absorption. (10)

OR

- VII. Derive an equation for the diffusion of a component A through a stagnant gas B under steady state conditions. How is the equation different if B also diffuses in the opposite direction (equimolar counter diffusion)? (10)

- VIII. (a) Define bound moisture, equilibrium moisture and free moisture. Explain the mechanism of drying with the help of rate of drying curve. (5)

- (b) Name the various equipments used for liquid-liquid extraction. (5)

OR

- IX. (a) Describe the procedure for obtaining the number of theoretical plates by McCabe Thiele method in a distillation column. (6)

- (b) Define relative volatility. Obtain an equation to describe vapour phase composition from relative volatility. (4)
